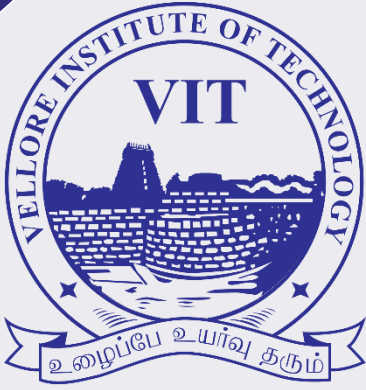




EMBEDDED SYSTEMS & IOT LAB

CHALLENGING TASK 4

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Vishva S A (23MIS0420)

QUESTION:-

Design a wearable health monitoring device for elderly patients that continuously measures and displays their heart rate using a Pulse Sensor and Arduino microcontroller.

The system should:

- Sense the heartbeat through a pulse sensor placed on the Fingertip.
- Process the sensor data to calculate the beats per minute (BPM).

Display the real-time heart rate on a Blynk application.

- Trigger a buzzer alert if the heart rate goes beyond a specified safe range (e.g., above 100 BPM or below 60 BPM).

```

#define BLYNK_PRINT Serial
#define BLYNK_TEMPLATE_ID "TMPL39IuX1m8Z"
#define BLYNK_TEMPLATE_NAME "Heart Monitor"
#define BLYNK_AUTH_TOKEN "7B3euQq0D6ElR79U2c8p-N3cQ92rAvvz"
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>

char auth[] = " BLYNK_AUTH_TOKEN ";
char ssid[] = "iot";
char pass[] = "password";

const int pulsePin = A0;
const int buzzer = 8;
const int led = 7;
int upperLimit = 100;
int lowerLimit = 60;
int prevBpm = 0;
BlynkTimer timer;

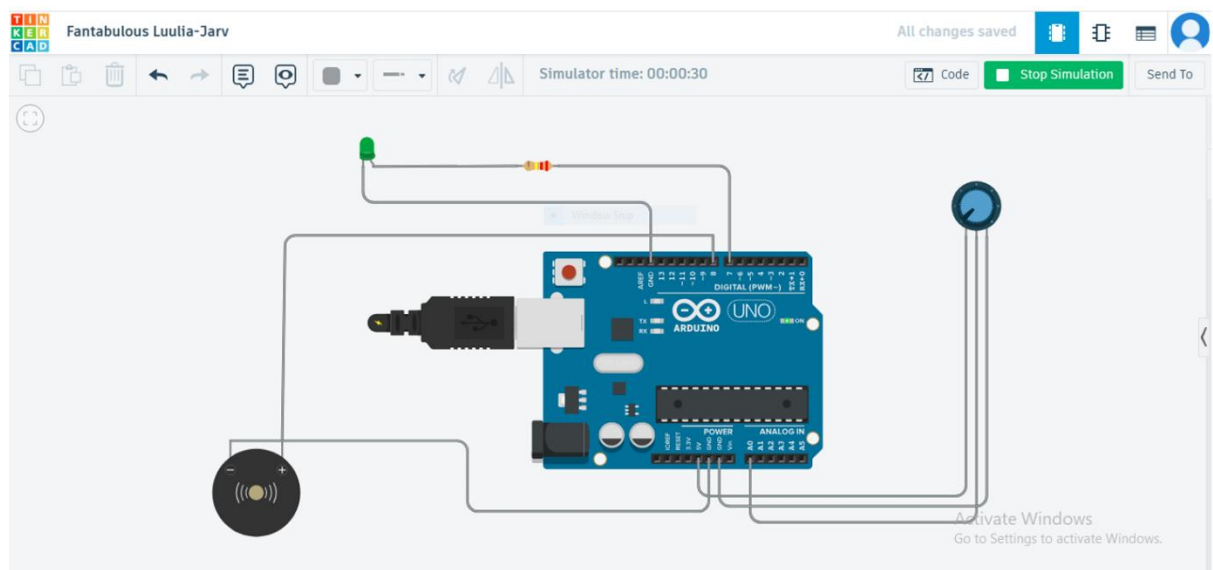
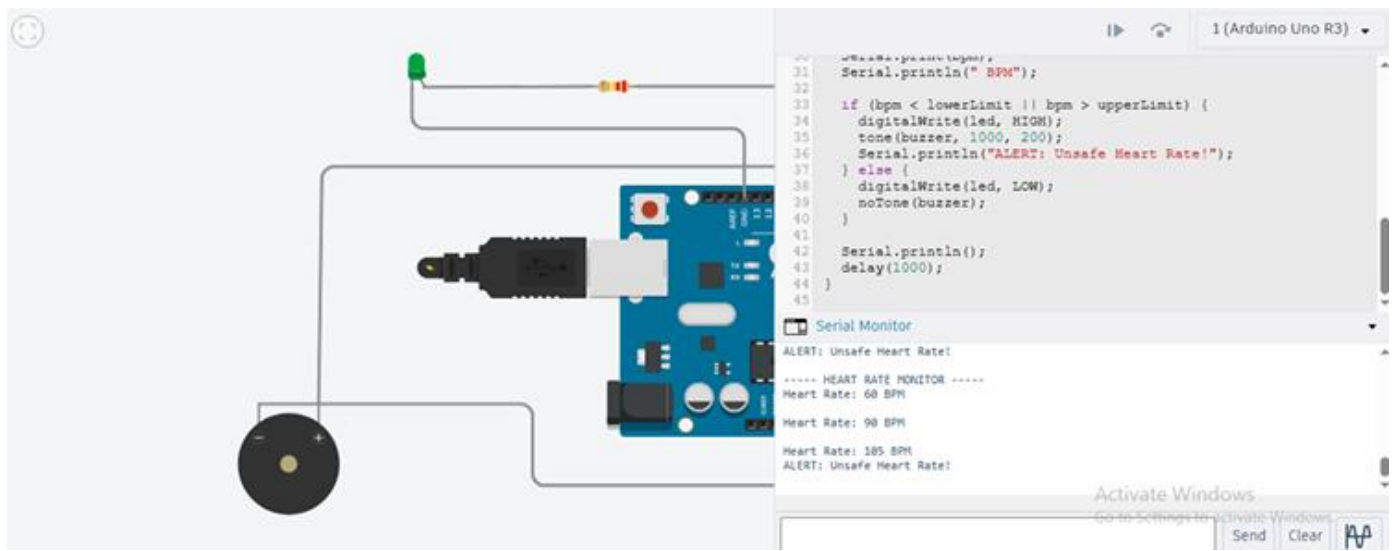
void sendHeartRate() {
    int sensorValue = analogRead(pulsePin);
    int bpm = map(sensorValue, 0, 1023, 50, 120);
    bpm = (bpm + prevBpm) / 2;
    prevBpm = bpm;

    Blynk.virtualWrite(V0, bpm);
    if (bpm < lowerLimit || bpm > upperLimit) {
        digitalWrite(led, HIGH);
        tone(buzzer, 1000, 200);
    }
    else {
        digitalWrite(led, LOW);
        noTone(buzzer);
    }
}

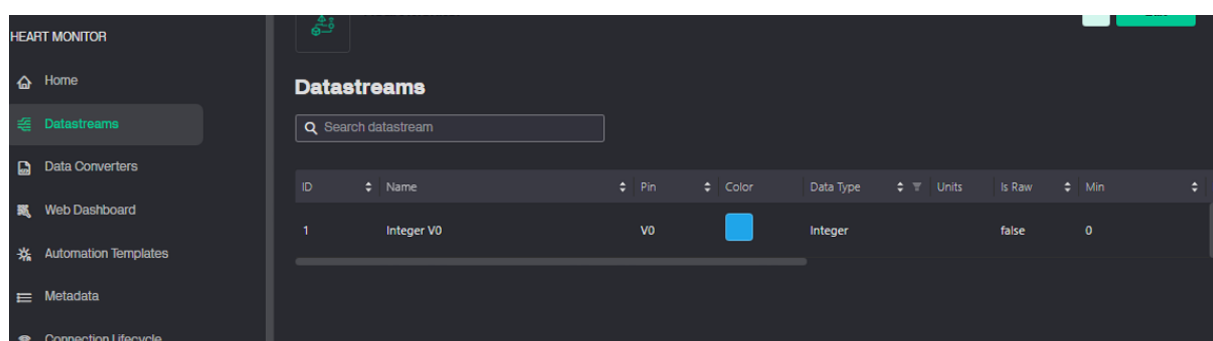
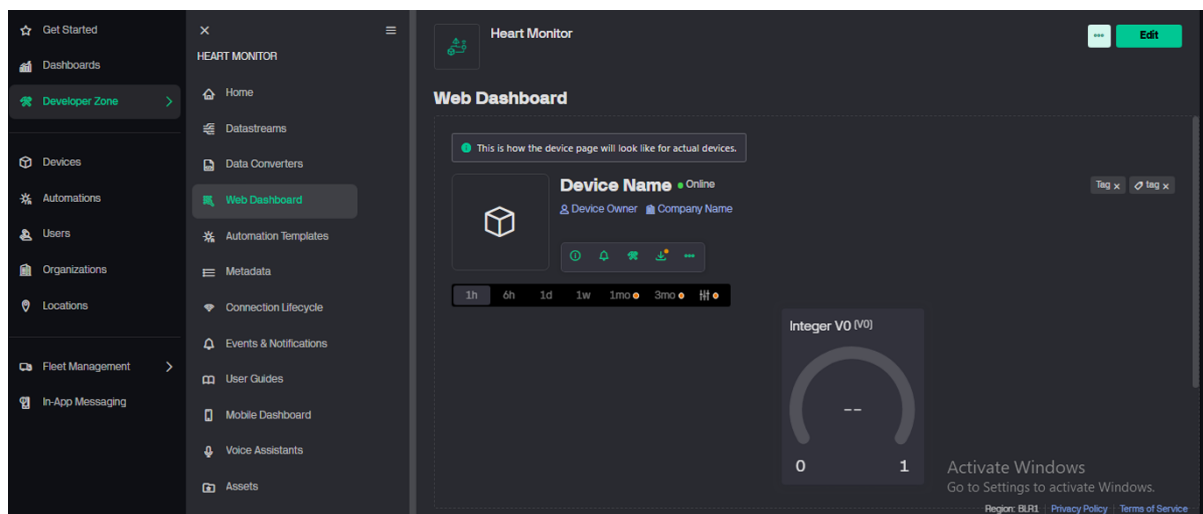
void setup() {
    Serial.begin(9600);
    Blynk.begin(auth, ssid, pass);
    pinMode(buzzer, OUTPUT);
    pinMode(led, OUTPUT);
    timer.setInterval(1000L, sendHeartRate);
}

void loop() {
    Blynk.run();
    timer.run();
}

```



Blynk output:



Code:

```
#define BLYNK_PRINT Serial
#define BLYNK_TEMPLATE_ID "TMPL39IuX1m8Z"
#define BLYNK_TEMPLATE_NAME "Heart Monitor"
#define BLYNK_AUTH_TOKEN "7B3euQq0D6ElR79U2c8p-N3cQ92rAvvz"

#include <WiFi.h>
#include <BlynkSimpleEsp32.h>

char ssid[] = "VISHVA";
char pass[] = "THALA_AJITH";
char auth[] = BLYNK_AUTH_TOKEN;

const int pulsePin = A0;
const int BUZZER_PIN = 8;
const int LED_PIN = 7;

const int LOWER_LIMIT = 60;
const int UPPER_LIMIT = 100;
const unsigned long SAMPLE_INTERVAL_MS = 10;
const unsigned long LED_BLINK_MS = 120;
const unsigned long BLYNK_SEND_INTERVAL_MS = 1000;
const int BPM_SMOOTHING_COUNT = 4;

float bpm = 0.0;
float bpmHistory[BPM_SMOOTHING_COUNT];
int bpmHistoryIdx = 0;
unsigned long lastSampleTime = 0;
unsigned long ledOffTime = 0;
unsigned long lastBlynkSend = 0;
bool buzzerOn = false;
int prevBpm = 0;

BlynkTimer timer;

void setBuzzer(bool on) {
    digitalWrite(BUZZER_PIN, on ? HIGH : LOW);
    buzzerOn = on;
}

float smoothBPM(float newBpm) {
    bpmHistory[bpmHistoryIdx] = newBpm;
    bpmHistoryIdx = (bpmHistoryIdx + 1) % BPM_SMOOTHING_COUNT;
    float sum = 0;
    int count = 0;
    for (int i = 0; i < BPM_SMOOTHING_COUNT; ++i) {
        if (bpmHistory[i] > 0) {
```

```

        sum += bpmHistory[i];
        count++;
    }
}
return (count == 0) ? newBpm : (sum / count);
}

void sendBPM() {
    if (millis() - lastBlynkSend < BLYNK_SEND_INTERVAL_MS) return;
    lastBlynkSend = millis();
    Blynk.virtualWrite(V0, (int)(bpm + 0.5));
}

void readHeartRate() {
    int sensorValue = analogRead(pulsePin);
    int rawBpm = map(sensorValue, 0, 1023, 50, 120);
    rawBpm = (rawBpm + prevBpm) / 2;
    prevBpm = rawBpm;
    bpm = smoothBPM(rawBpm);
    digitalWrite(LED_PIN, HIGH);
    ledOffTime = millis() + LED_BLINK_MS;
    if (bpm < LOWER_LIMIT || bpm > UPPER_LIMIT) {
        setBuzzer(true);
    } else {
        setBuzzer(false);
    }
    sendBPM();
}

void setup() {
    Serial.begin(9600);
    pinMode(BUZZER_PIN, OUTPUT);
    pinMode(LED_PIN, OUTPUT);
    digitalWrite(BUZZER_PIN, LOW);
    digitalWrite(LED_PIN, LOW);
    for (int i = 0; i < BPM_SMOOTHING_COUNT; ++i) bpmHistory[i] = 0;
    Blynk.begin(auth, ssid, pass);
    timer.setInterval(1000L, sendBPM);
    Serial.println("Heart Monitor Started");
}

void loop() {
    Blynk.run();
    timer.run();
    unsigned long now = millis();
    if (ledOffTime != 0 && now >= ledOffTime) {
        digitalWrite(LED_PIN, LOW);
        ledOffTime = 0;
    }
}

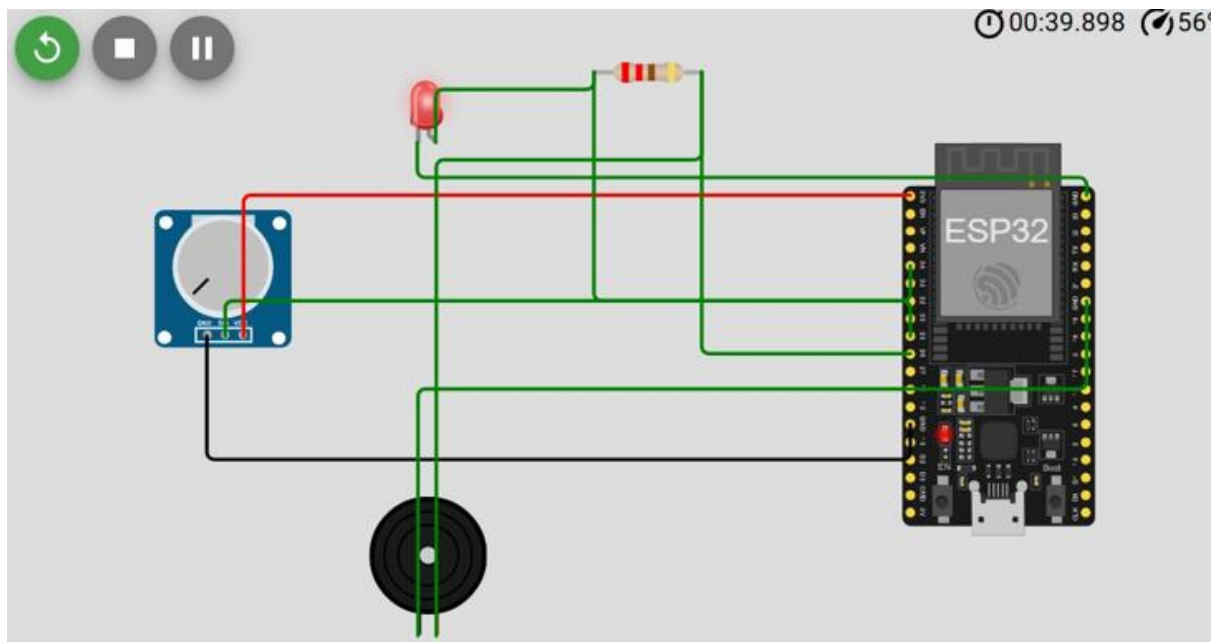
```

```

}
if (now - lastSampleTime >= SAMPLE_INTERVAL_MS) {
    lastSampleTime = now;
    readHeartRate();
}
}

```

Output:



```

rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:2
load:0x3fff0030,len:1156
load:0x40078000,len:11456
ho 0 tail 12 room 4
load:0x40080400,len:2972
entry 0x400805dc
Heart monitor started
BPM: 75
BPM: 79
BPM: 77
BPM: 77
BPM: 74
BPM: 75
BPM: 77

```